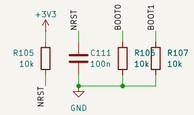
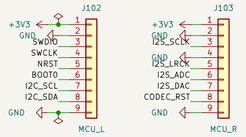
BIG MODULAR USB-AUDIO — MCU board, S1. CH32V307RCT6, USB 2.0 High-Speed (UAC2, 192 kHz / 24-bit, 2 in / 2 out). Circuit intent: docs/research.md

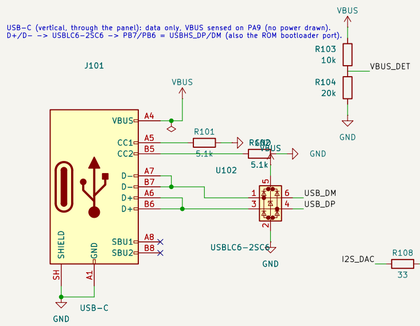
BOOT/RESET buttons and the SWD header are on the carrier (B side).
Hold BOOT, press RESET: ROM bootloader on the USB-C (wchisp / WCHISPTool).

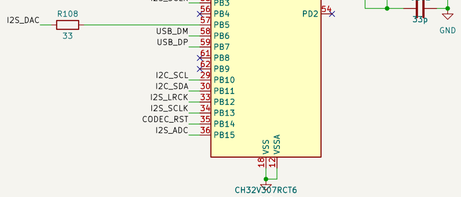



CARRIER HEADERS: J102 = carrier J6, J103 = carrier J7 (2.54 mm pin strips,
soldered through both boards; the strips are in the carrier BOM).

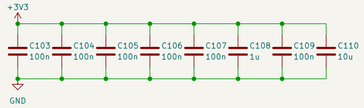



USB-C (vertical, through the panel): data only, VBUS sensed on PA9 (no power drawn).
D+/D- -> USBLC6-2SC6 -> PB7/PB6 = USBHS_DP/DM (also the ROM bootloader port).




I2S2 slave RX: PB13 CK, PB12 WS, PB15 SD (codec SDOUT). I2S3 slave TX: PB3 CK, PA15 WS,
PB5 SD (codec SDIN). I2C2: PB10/PB11. PB14 codec RST. The codec is the clock master.




Circuit intent: docs/research.md; placement: docs/geometry.json
SCHEMATIC S1 — CH32V307RCT6 USB 2.0 High-Speed (UAC2), vertical USB-C through the panel; sold

| | | |
|---|---|---|
| **Big Modular** | | |
| Sheet: / | | |
| File: usb-audio-mcu.kicad_sch | | |
| ***Title: Big Modular — USB-AUDIO MCU board*** | | |
| Size: A2 | Date: | **Rev: S1** |
| KiCad E.D.A. 10.0.6 | | Id: 1/1 |